

PalmSource®

Conference and Expo



WebSphere Mobilizes Your Enterprise

The right platform ...and the right tools

Paul Buck, President and CEO
Object Technology International, Inc.
A subsidiary of IBM



How does IBM optimize Palm OS® handhelds for the deployment of e-business applications in a post-PC era?



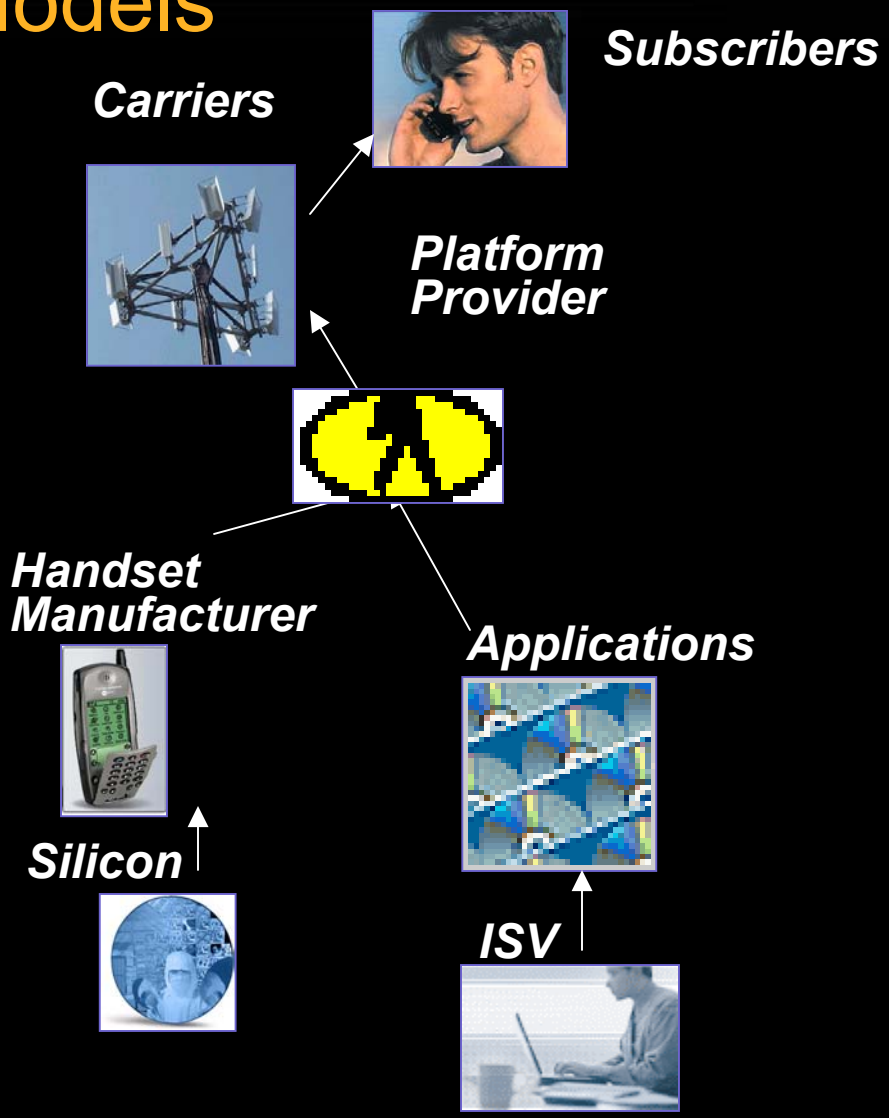
IBM Strategy - Mobilize the Enterprise

- Extend e-business applications to Palm OS® devices in a post-PC era
  - PDAs, smartphones, hand-helds
- Build an end-to-end device to services solution
  - Data services, transaction services, application services
- Drive compelling device-to-services programming model
  - Break through device barrier with end-to-end solutions
  - Break through “browser mentality” with “semi-connected” devices
  - PIM becomes EIM: e-business information management
- Combine the power of WebSphere™ with convenience of Palm OS devices

Wireless e-business Models

Catalogue of services
- 1990s call / browse
- 2000s e-mail / SMS
- 2002+ e-business

e-business extends existing:
- Data - synchronization
- Transactions / assured msgs
- Application - access

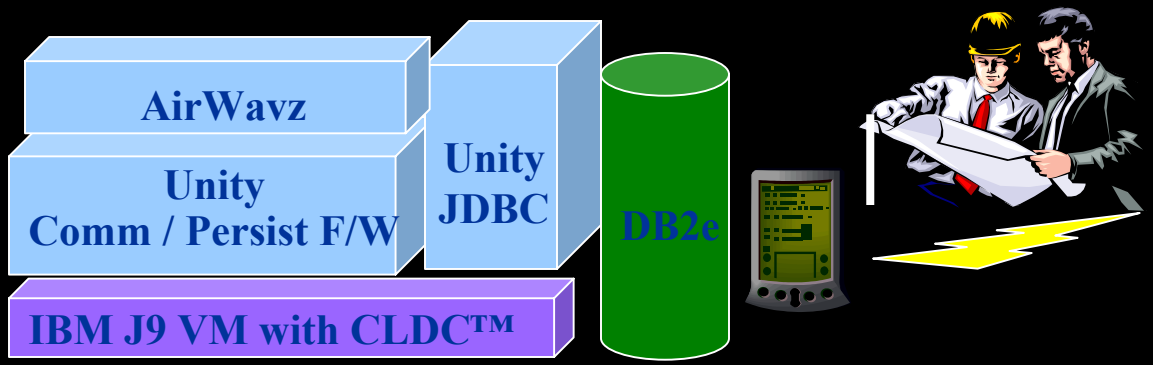


Mobile Customer Solutions - **Van Gend & Loos**

- Van Gend & Loos - Euro Express
- Business Problem
  - Tracking and tracing of package deliveries
    - administration of deliveries for driver
    - real-time status to office / customer
- Solution
  - Symbol Device scan bar codes on package
  - Package status / destination stored on device
    - Java(TM) application – IBM J9 virtual machine on Palm OS
    - UI framework developed by Compuware
    - Communication via RF Lan and Mobitex WAN
  - Integration provided by IC Group – www.icgroup.nl

Mobile e-business Solutions - **AirToolz**

- Business Problem - Residential Construction Schedules
  - One change affects many parties
  - Superintendents, contractors and home office
- Solution - Central Schedule on WebSphere
  - Superintendent updates using Wireless Palm - J9 virtual machine
  - Subcontractors affected immediately notified
  - Managers instantly see delays via browser
  - Integration provided by Unity Software - www.unity-software.com
- Bottom Line
  - Real-time communications streamlines the building process



Mobile Wireless Solutions - **Applied Reasoning**

- Business Problem
  - Instant access to IT assets
  - Overcome bandwidth issues
- Solution - Mobile Classic Blend
  - Smart Client with J9 virtual machine
    - Kyocera smartphone (Palm OS)
  - Middleware minimizes client / server interaction
    - Only vital data / messages sent
  - Instant response / instant access to messages / IT assets
- Why IBM?
  - "IBM's **J9** was chosen because of its high performance, small footprint and integration with the Palm operating system," said William Burdick, Applied Reasoning's Executive Vice President and Chief Scientist. "VisualAge Micro Edition provides superior tool support which was critical to our aggressive development cycle."



WebSphere e-business - The Right Platform

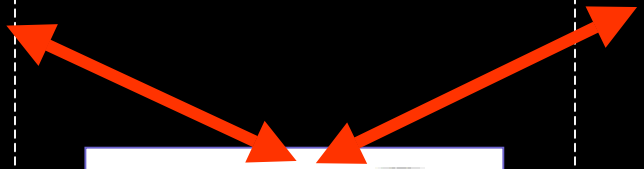
- *Device to Services* - Entertainment/Information/e-business
  - It is not about browsing, it is about e-business
  - Applications / Data / Transactions
- *Time to Market* – respond quickly to meet demand
  - Product Level Platform - tested, certified, supported
- *Footprint and Speed* - fast, compact and tuned for each device
  - Proven in independent testing

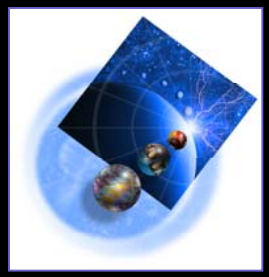
Wireless World is Headed to end-to-end e-business

J2ME™







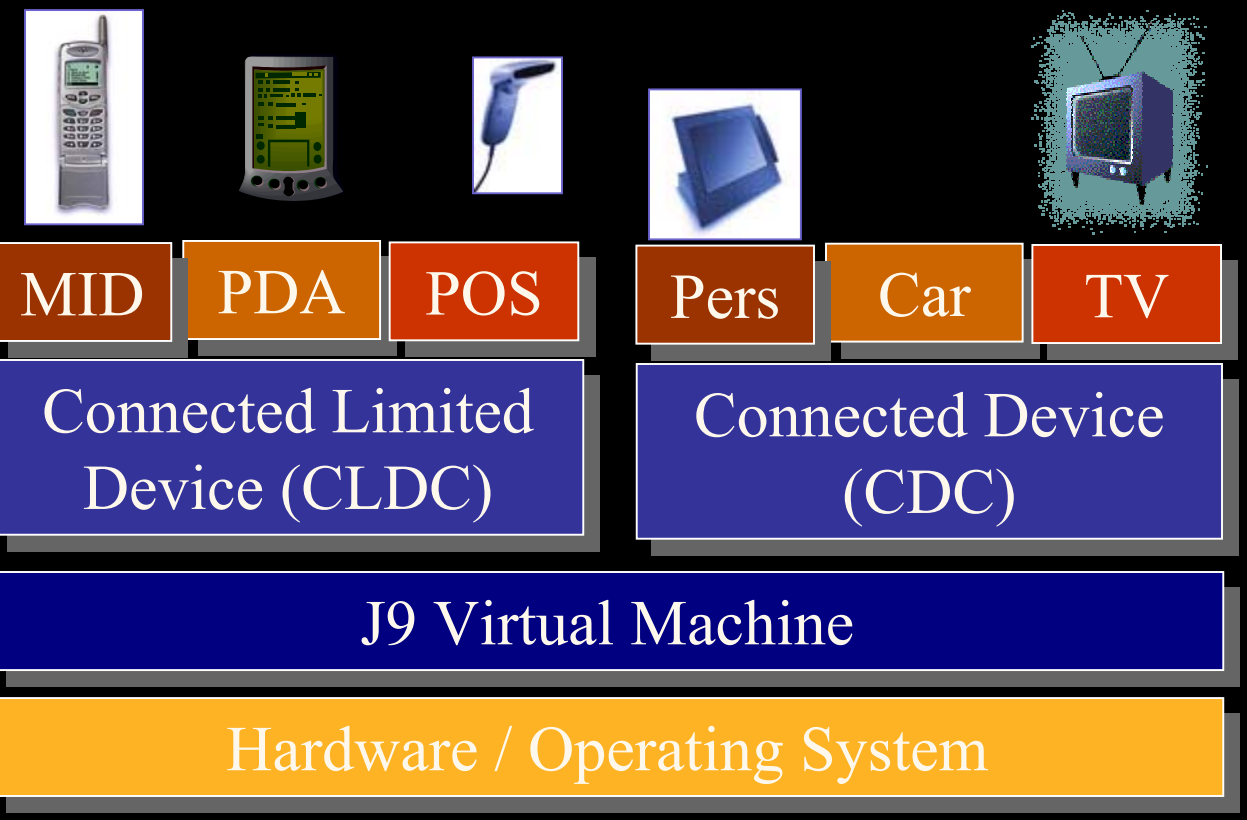


VisualAge Micro Edition

Develop
Debug
Deploy
Maintain

Download at www.ibm.com/embedded

Java 2 Platform, Micro Edition (J2ME)



J2ME promises platform portability

The Small, Fast, Flexible J9 Virtual Machine

- J9 VM, a Java virtual machine that meets or exceeds industry standards
  - J9 is Java Powered!
    - Supports the J2ME CLDC, MIDP, CDC and Foundation
  - VM is customized for Palm OS on the Motorola 68K processor
  - Executes in an extremely small memory space
  - Same technology available for other platforms like Windows, Linux and QNX Neutrino
    - Palm OS shares performance enhancements and fixes with other supported platforms
  - Modular architecture lets developers configure the VM to include or exclude certain components
- J9 is delivering on ARM® today for Linux, OSE, and QNX
  - Poised for delivery on Palm OS

VisualAge® Micro Edition

- Award-winning tools for embedded Java development
- Optimize for space
  - SmartLinking technology
  - Remove unneeded classes, methods and fields
- Optimize for speed
  - Just-in-time (JIT) and ahead-of-time (AOT) compilation technology
  - Precompile those methods that matter most
- “On Target” execution analysis
  - Remote debug of application live on the device
  - MicroAnalyzer to track method calls, memory usage and thread switches
  - Find and eliminate the bottlenecks
- Moving to IBM WebSphere Workbench Technology

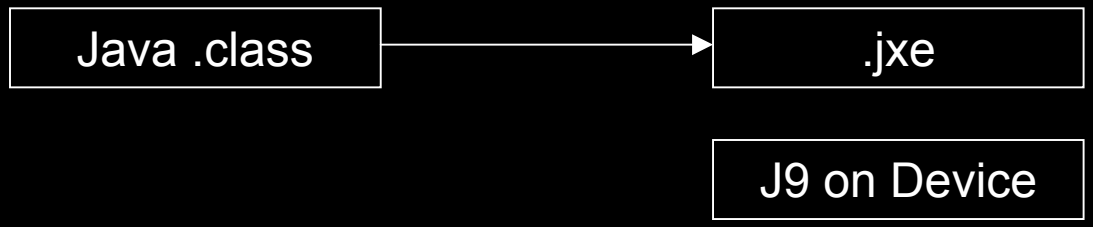


Many Jobs, One Interface

- From the single interface presented by the IDE you can:
  - Edit and search
  - Build an executable image
  - Prepare the image for ROM
  - Load the target
  - Perform full source code-level debugging
  - Replace code on the fly
  - Analyze and profile at near real-time execution speed
- Team programming made simple and resourceful
- Automated build to generate a Palm PRC from Java source files

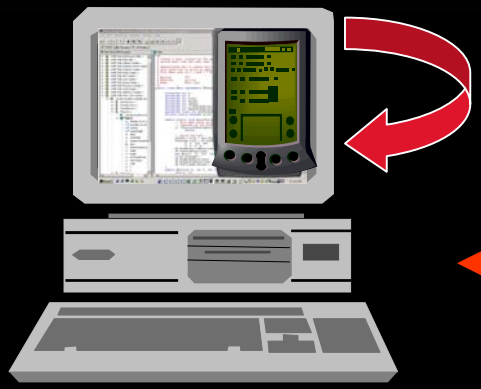
Streamline for Speed - SmartLinker

- "Pre-Links" class files for execution on the target device



- Increases performance
  - Pre-processes code for faster loading / execution
  - Execute in place (XIP) for "instant on" experience
  - Optionally in-lines methods
  - Reduces the overhead of dynamically linking class files
- Use memory on your Palm device efficiently
  - Removes unneeded classes, methods, fields

Remote Debug Tests as Application Runs “Live”



Local Debug

via emulator

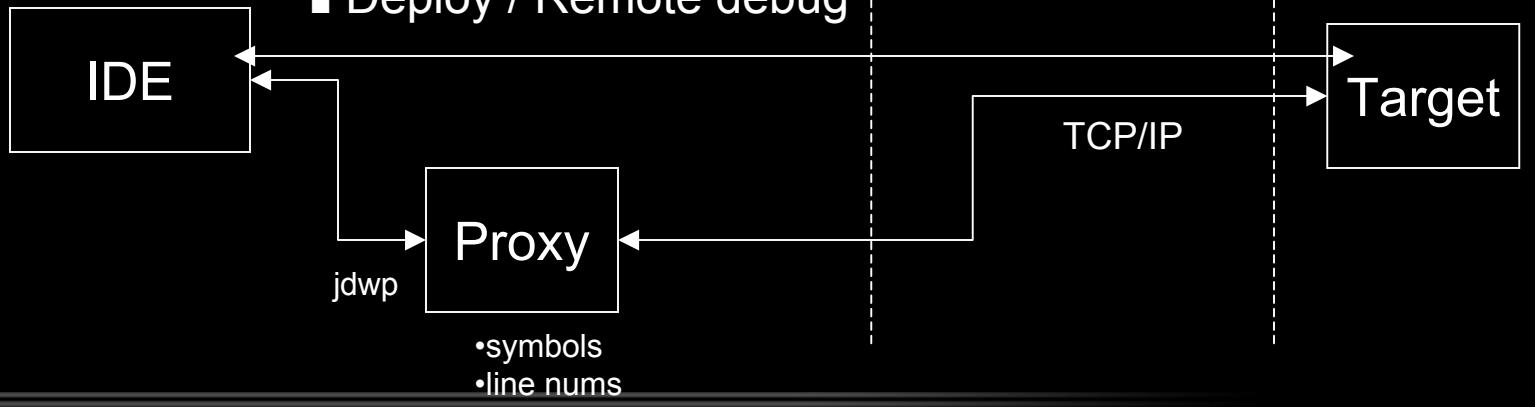
Remote Debug

“Live on target”



Integrated Development Environment

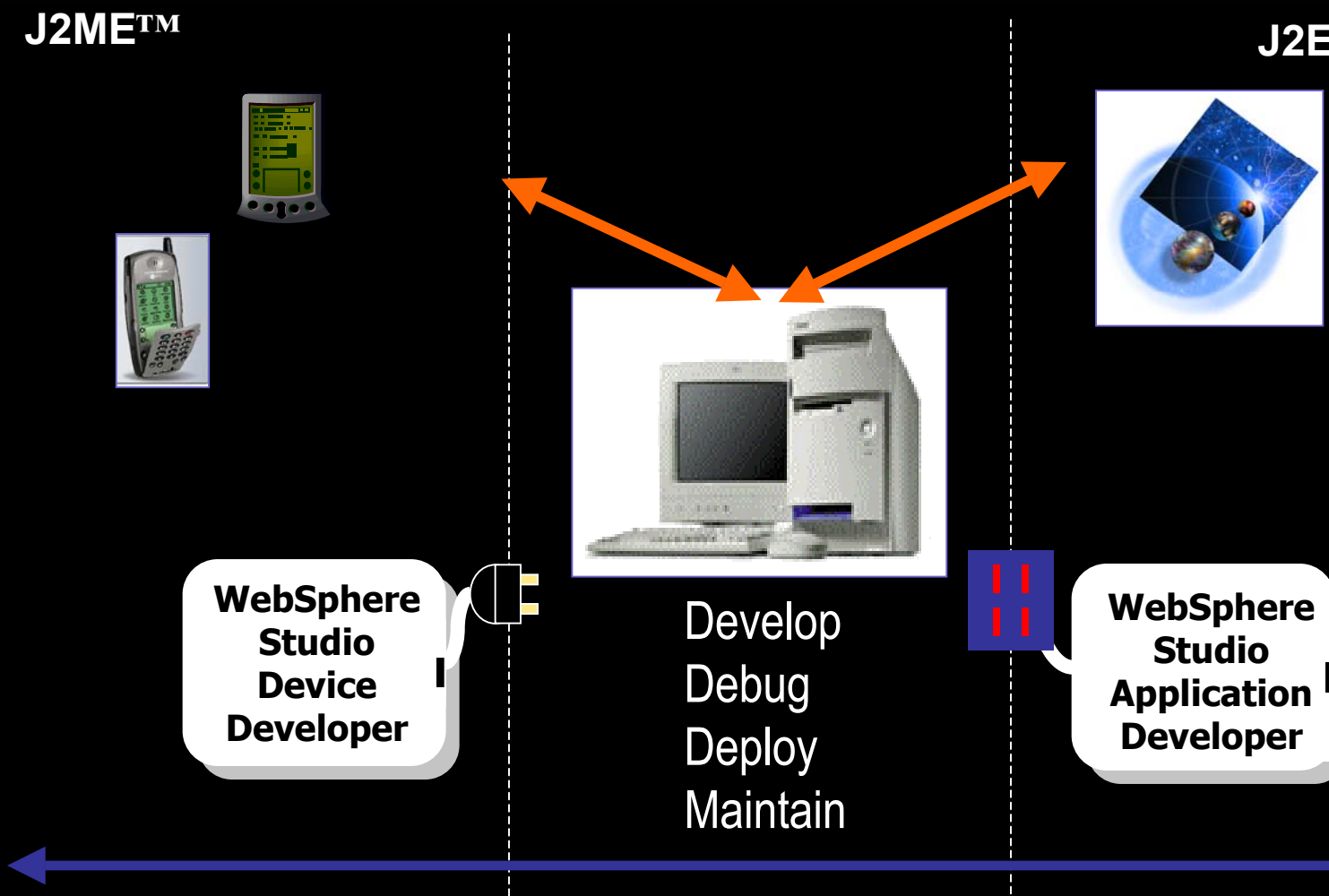
- Edit, compile, link, build
- Local test (emulator)
- Deploy / Remote debug



Making Java Applications Run Efficiently

- MicroAnalyzer helps you streamline, speed up, and pare down your Java application
  - MicroAnalyzer can operate on a device with minimal overhead
  - Provides snapshot data captured from the VM
    - Identify wasted space in thread stacks
    - Fine-tune garbage collection by the VM
  - Real-time tracing of application execution
    - Debug thread contention and deadlocks
    - Instrument your code with user-defined events
    - Watch your application run “live”
- When you really need to know what your application is doing, the MicroAnalyzer provides the answers
- Available for ARM based platforms

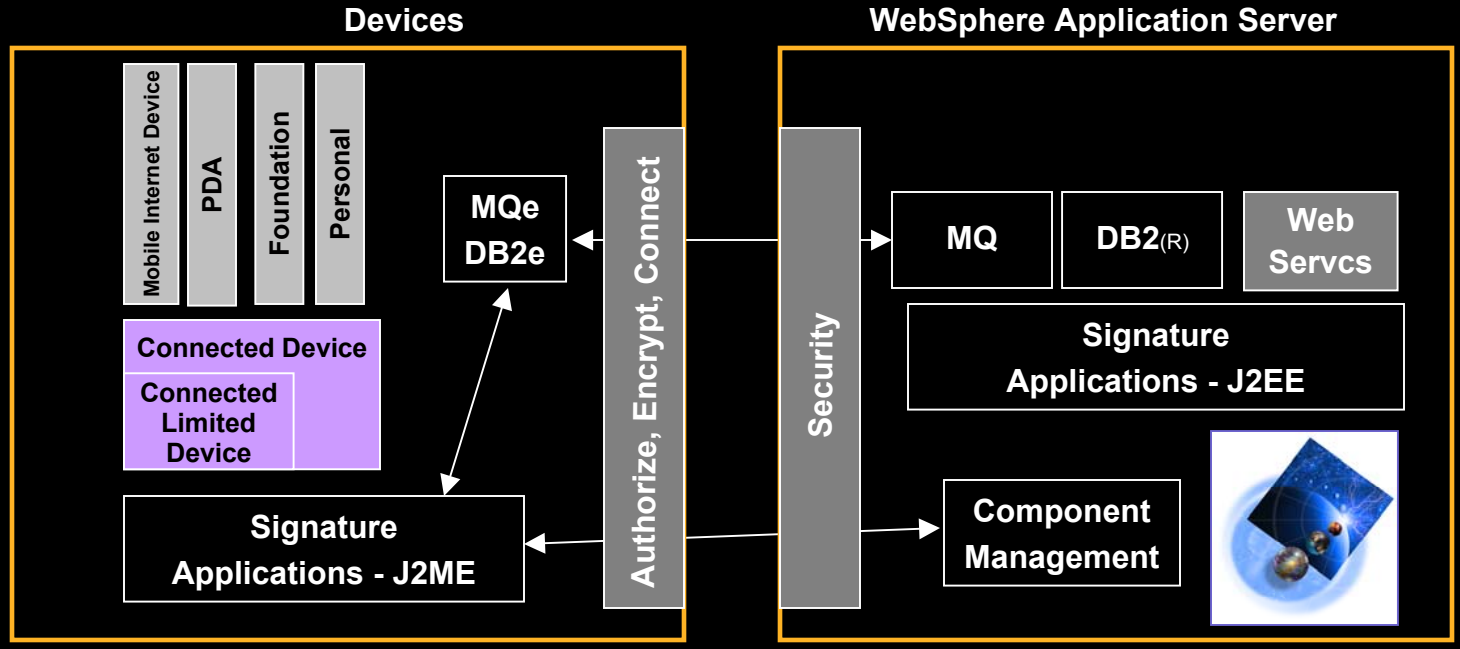
The Vision Becomes Reality: end-to-end e-business



Beta at www.ibm.com/embedded

The Right Platform - Mobilize Your Enterprise

- End-to-end device to services - J2ME meets J2EE



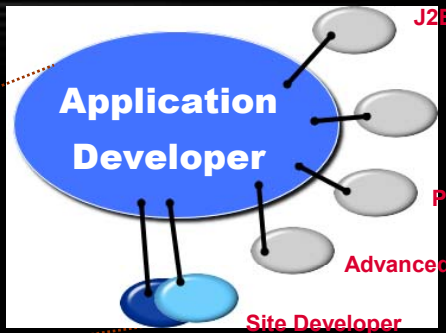
Scenario: Mobile e-business - one time message

- Messaging Services - WebSphere MQ
- Data Services - DB2e, Cloudscape
- The Power of WebSphere, the convenience of the device

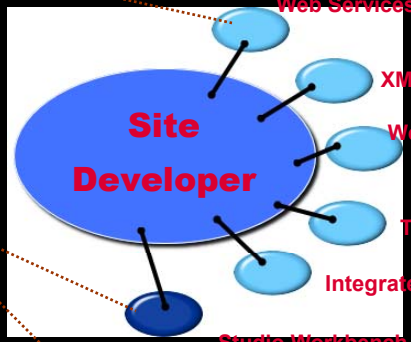
The Right Tools



J2EE(TM)



Site Dev't



Contribute

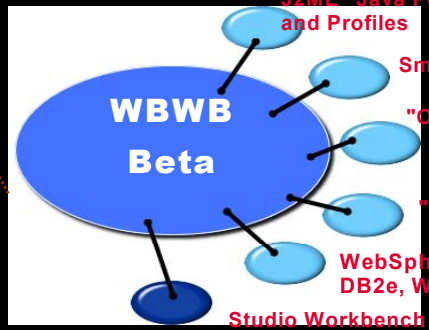




Adopt



J2ME



IBM Commitment to Palm OS Devices

- Data Services
  - Lotus Notes - collaborative data
  - DB2e - Relational Data (Java technology)
  - Cloudscape (DB2j) - Object Data
- Transaction Services
  - MQe – WebSphere MQ everyplace - one time messaging
- Application Services
  - IBM J9 Virtual Machine - Runtime environment for Java applications

WebSphere e-business Platform

- Growing developer community - ibm.com/developerworks
  - 600,000 registered developers, 5,000 Trained IBM Professionals
  - 150 WW user groups
  - 300K downloads, 20Million web hits / year
- Growing e-business leadership - ibm.com/websphere
  - 10X Rate of nearest competitor
  - 10 consecutive quarters of double digit growth
  - 75% YtY (3Q/2001) increase in Revenue
- Growing leadership in application development - ibm.com/embedded
  - Market Leader in developing Java applications for the enterprise
  - 2 consecutive years of double digit grown
  - Over 50 industry awards for IDE



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